AN INTRODUCTION TO THE PCA'S OPTIONAL RULES FOR ARBITRATION OF DISPUTES RELATING TO OUTER SPACE ACTIVITIES

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I. INTRODUCTION

On December 6, 2011, the Administrative Council of the Permanent Court of Arbitration (PCA) adopted the *Optional Rules for Arbitration of Disputes Relating to Outer Space Activities* (Outer Space Rules).[1] In an effort to address fundamental *lacunae* in the existing dispute resolution mechanisms of international space law, the Outer Space Rules were created to provide a means of voluntary and binding dispute resolution available to all parties engaged in outer space activities and tailored to the particularities of this unique area of economic activity. In the few months since their completion, they have already attracted attention from legal practitioners representing actors in outer space activities.

The Outer Space Rules are the product of over two years of dedicated work by a group of international experts, in conjunction with the International Bureau of the PCA. Having had the pleasure to chair the expert group throughout, I propose to introduce the Outer Space Rules to the academic community by

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[1] Optional Rules for Arbitration of Disputes Relating to Outer Space Activities, Dec. 6, 2011, *available at* http://www.pca-cpa.org/upload/files/Outer%20Space%20Rules.pdf.

providing insight into the factual and intellectual processes of their development.[2]

II. THE INITIAL STAGES

The initiative for the development of a set of specialized arbitral rules for disputes that arise in the space sector came from the PCA, an intergovernmental organization counting 115 member states.[3] One of the PCA's principal functions is to facilitate dispute resolution, including arbitration, between various combinations of states, state entities, intergovernmental organizations, and private parties. The PCA's secretariat, the International Bureau, headed by the PCA Secretary-General, provides full registry services and legal and administrative support to arbitral tribunals and commissions. Most significantly for our purposes, since 1992 the PCA has adopted eight sets of party and sector-specific rules of procedure for arbitration or conciliation developed by expert groups. In 2009, inspired by these experiences and suspecting the lack of an adequate dispute resolution mechanism for space-related disputes, the Administrative Council of the PCA approved the establishment of an Advisory Group of legal experts ("Advisory Group"), with a mandate to:

> ... firstly ... assess generally the need for a final and binding dispute-settlement mechanism for disputes involving the use of outer space by states, intergovernmental organizations and private entities and, specifically, ... highlight the benefits of arbitration in this regard. Secondly, the Advisory Group will draw up optional rules to this end for inclusion in the PCA's set of arbitration rules.[4]

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[2] In parallel to this text, another introduction to the Outer Space Rules by Dr. Stephen Hobe is being published (in English) in the 1 *Zeitschrift für Luft- und Weltraumrecht* (German Journal of Air and Space Law) (2012).

[3] As of June 2012. *See* Permanent Court of Arbitration, *Member States*, http://www.pca-cpa.org/showpage.asp?pag_id=1038 (last visited June 3, 2012).

[4] Letter from the PCA Secretary-General to the members of the PCA Administrative Council (May 29, 2009) (on file with the PCA).

The then Secretary-General of the PCA, Mr. Christiaan M.J. Kröner, officially invited me to chair the Advisory Group on November 17, 2009.

My first mission as Chair was to compose an Advisory Group on the basis of the highest internationally acknowledged professional qualifications, with due regard given to achieving a broad geo-political representation, reflective of the PCA's wide state membership. The Advisory Group's substantive knowledge of space law, including its dispute resolution aspect, was to complement the PCA International Bureau's extensive practical experience in the use of various sets of procedural rules in arbitral proceedings. Accordingly, the members of the Advisory Group were: Dr. Tare Brisibe (Nigeria), Prof. Frans von der Dunk (Netherlands), Prof. Joanne Gabrynowicz (United States), Prof. Dr. Stephan Hobe (Germany), Dr. Ram Jakhu (Canada), Prof. Armel Kerrest (France), Mrs. Justine Limpitlaw (South Africa), Prof. Dr. Francis Lyall (United Kingdom), Prof. V.S. Mani (India), Mr. Jose Montserrat Filho (Brasil), Prof. Dr. Maureen Williams (United Kingdom/Argentina), and Prof. Haifeng Zhao (China).

III. ARBITRATION AS A SECTORIALIZED DISPUTE RESOLUTION MECHANISM FOR SPACE LAW

Upon constitution, the Advisory Group directly embarked on fulfilling the first part of its mandate: to consider the desirability of, or need for, arbitration rules specifically targeted at the resolution of space-related disputes. This phase was conducted through questionnaires, multiple rounds of comments, and a survey of existing outer space related instruments that either contained or could have contained provisions for dispute resolution through arbitration. When these documents revealed a general consensus in support of arbitration among the Advisory Group, I resolved to take advantage of the highly specialized expertise of the Advisory Group members, and invited those members who were willing to submit discussion papers on the dispute resolution needs of particular areas of space law.

Five discussion papers were submitted and circulated within the Advisory Group.[5]

The Advisory Group's starting-point for these discussions was that in the last twenty-five years a relatively firm consensus seems to have emerged in academia affirming the need for a sectorialized dispute resolution mechanism for disputes relating to outer space activities. Proposals have been made for a variety of solutions, including the establishment of a new international court for space law.[6] Given the existing scholarship on this subject, we chose not to dwell on all aspects of dispute resolution in space law, but to focus on: (1) noting the relevant contemporary characteristics of outer space activities, (2) evaluating whether arbitration could provide an effective means for dispute resolution in an area possessing such characteristics, and (3) devising how existing procedural rules for arbitration could be modified to better fit the particularities of space-related disputes.

We first noted that the past few decades have seen a steady rise in space-related activity, primarily due to an increase in the commercial uses of outer space, especially in the sectors of satellite communications, launching services, and remote sensing.[7] It seems reasonable to suppose that this increase in activity augments the risk of disputes.[8]

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[5] Ram S. Jakhu, *Dispute Resolution under the ITU Agreements* (Advisory Group Discussion Paper, 2010); Frans G. von der Dunk, *Private Commercial Manned Spaceflight and Dispute Settlement* (Advisory Group Discussion Paper, 2010) [hereinafter von der Dunk, Discussion Paper]; Maureen Williams, *Satellite Data and its value as evidence in international litigation* (Advisory Group Discussion Paper, 2010) [hereinafter Williams, Discussion Paper]; Tare Brisibe, *Alternative Dispute Resolution of International Investment Disputes in Public-Private Space Projects* (Advisory Group Discussion Paper, 2010) [hereinafter Brisibe, Discussion Paper]; Joanne Irene Gabrynowicz, *Remote Sensing and Potential Optional Rules for Arbitration of Disputes Relating to Outer Space* (Advisory Group Discussion Paper, 2010) [hereinafter Gabrynowicz, Discussion Paper] (on file with the PCA).

[6] *See e.g.*, Final Draft of the Revised Convention on the Settlement of Disputes Related to Space Activities, at art. 37, *as amended in* REPORT OF THE SIXTY-EIGHTH CONFERENCE OF THE ILA (Taipei, 1998) (hereinafter 1998 TAIPEI CONFERENCE REPORT).

[7] Chia-Jui Cheng, *International Arbitration System as Mechanism for the Settlement of Disputes Arising in Relation to Space Commercialization*, 5 SINGAPORE J. INT'L & COMP. L. 167 (2001).

[8] Maureen Williams, Rapporteur, *Space Committee of the International Law Association*, *in* 1998 TAIPEI CONFERENCE REPORT, *supra* note 6, at 241. A tendency toward dispute avoidance in the space sector must however be noted (Brisibe, Discussion Paper, *supra* note 5, at 11).

There has likewise been an increase in the number and variety of the actors involved in space activities. In a field long dominated by the U.S.A. and the former U.S.S.R., there are now over thirty countries possessing significant space industries.[9] Moreover, there has been a notable relaxation of government control on space activities.[10] This factor, coupled with an increase in the possible commercial uses of outer space, has led to the influx of a variety of non-state actors onto the stage of space law. Space-related disputes can now arise between states, state agencies, intergovernmental regional or international organizations, and private entities, such as national and multinational corporations.[11] With the advent of space tourism, even private persons may become entangled in disputes relating to outer space activities.[12]

Further, due to the high level of financial and scientific investment required by most uses of outer space, the space sector exhibits a high level of international cooperation, both between states and private entities of various nationalities.[13] Thus a great number of space-related disputes are likely to arise at an international level.

An effective dispute resolution mechanism in space law would therefore be international, accessible to a variety of public and private parties, and capable of responding to potentially high demand for dispute resolution. In this regard, the Advisory Group noted that existing dispute resolution mechanisms in international space law present several *lacunae*, some of which merit mention.

In particular, numerous existing dispute resolution mechanisms are limited either in their personal or material scope. Thus, many mechanisms are not available to private parties.

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[9] GERARDINE MEISHAN GOH, DISPUTE SETTLEMENT IN INTERNATIONAL SPACE LAW: A MULTI-DOOR COURTHOUSE FOR OUTER SPACE 164 (Leiden, Boston, Martinus Nijhoff, 2007).

[10] *Id.* at 157.

[11] Frans G. von der Dunk, *Space for Dispute Settlement Mechanisms – Dispute Resolution Mechanisms for Space? A few legal considerations* (2001) *in* SPACE AND TELECOMMUNICATIONS LAW PROGRAM FACULTY PUBLICATIONS (2001), http://digitalcommons.unl.edu/spacelaw/38.

[12] von der Dunk, Discussion Paper, *supra* note 5, at 6-7.

[13] GOH, *supra* note 9, at 149-152.

International space law being initially conceptualized as a branch of public international law, dispute resolution was envisaged only as between states.[14] Moreover, even the most elaborate dispute resolution procedures for state-to-state disputes, found in the 1972 *Convention on International Liability for Damage Caused by Space Objects* (Liability Convention), suffer from a limited material scope (covering only claims for compensation for damage caused by space objects), – and a lack of binding power in the absence of specific agreement by the parties.[15] None of the other fundamental space law treaties provide any specific guidance as to dispute resolution. The founding convention of the European Space Agency (ESA) provides for arbitration between two or more member States, or between States and the agency, but of course only for disputes related to its interpretation and implementation.[16] The instruments of the International Telecommunications Union also provide for arbitration, but only as regards certain subject matters, such as harmful interference to registered radio frequencies.[17] In the absence of a specialized dispute resolution mechanism, States can rely on general mechanisms of dispute resolution available in public international law, for instance diplomatic negotiation or adjudication before the International Court of Justice.[18] However, these methods and venues are not available to private parties.

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[14] Cheng, *supra* note 7, at 165-66.

[15] Convention on International Liability for Damage Caused by Space Objects, *opened for signature* Mar. 29, 1972, 24 U.S.T. 2389, 961 U.N.T.S. 187; GOH, *supra* note 9, at 76.

[16] Convention for the Establishment of a European Space Agency, art. XVII, May 30, 1975, 14 I.L.M. 864. The ESA also includes arbitration clauses in all its external contracts. *Id.* at art. XXV.

[17] Constitution of the International Telecommunication Union, Dec. 22, 1992, 1825 U.N.T.S. 31251; Final Acts of the Additional plenipotentiary Conference, APP-92, at 71 (1992), *available at* http://www.itu.int/pub/S-CONF-ACTF-1992; Optional Protocol on the Compulsory Settlement of Disputes Relating to the Constitution of the International Telecommunication Union, to the Convention of the International Telecommunication Union and to the Administrative Regulations (1992), *available at* http://www.itu.int/net/about/basic-texts/optional-protocol.aspx.

[18] *See* Statute of the International Court of Justice, art. 34(1). June 26, 1945, 156 U.N.T.S. 77; 59 Stat. 1031.

Private parties may be inclined to resort to international commercial arbitration.[19] At the moment, international space law arbitration agreements between private parties generally provide for arbitration under the United Nations Commission on International Trade Law Arbitration Rules (UNCITRAL Rules) or the procedural rules of private arbitration institutions.[20] These rules, however, praised for being applicable to "the circumstances of various types of disputes and procedures," are not necessarily adapted to space-related disputes.[21]

Having considered the contemporary characteristics of outer space activities, the Advisory Group concluded that international arbitration has multiple advantages for the resolution of space-related disputes. First, arbitration is a method of dispute resolution open to all parties active in the field. The PCA, for instance, administers arbitrations that involve states, state-controlled entities, intergovernmental organizations, and private parties.

Second, arbitration is a voluntary mechanism, premised only on the consent of the parties.[22] This consent can be provided before a dispute arises by insertion of an arbitration clause in the legal instrument that defines the parties' relationship. In space law, this instrument can be an inter-State treaty, an agreement between a State and the space industry, or a commercial space contract between private enterprises or a private enterprise and a State agency.[23] The voluntary – or, as expressed in the title of the Outer Space Rules – "optional" aspect of arbitration is important where States are involved, as they may be more willing to agree to binding dispute resolution under discrete agreements than to enter into a new significant

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[19] FRANCIS LYALL & PAUL B. LARSEN, SPACE LAW: A TREATISE (Farnham, Ashgate, 2009); Brisibe, Discussion Paper, *supra* note 5, at 7.

[20] *See* International Chamber of Commerce Rules of Arbitration (2012), *available at* http://www.iccwbo.org/court/arbitration/id4199/index.html; or the London Court of International Arbitration Rules (1998), *available at* http://www.lcia.org/Dispute_Resolution_Services/LCIA_Arbitration_Rules.aspx; see Brisibe, Discussion Paper, *supra* note 5, at 11.

[21] THOMAS H. WEBSTER, HANDBOOK OF UNCITRAL ARBITRATION 4 (London, Sweet & Maxwell, 2010).

[22] *See* Outer Space Rules, *supra* note 1, at art. 1.

[23] Cheng, *supra* note 7, at 166.

multilateral treaty to the effect that all space-related disputes are to be dealt with in one way, be it, for example, through the establishment of an international court for space law, or the creation of an additional chamber to the International Court of Justice.[24]

Third, arbitration results in final and binding decisions.[25] This contrasts with the recommendatory nature of decisions under the Liability Convention. In arbitration, no appeals are possible and only limited grounds for challenge are available.[26] This can be of great importance given that space activities often operate on precise and fixed schedules, especially as regards the time windows for landing, atmospheric re-entry, descent and landing, and orbit insertion.[27] In these situations, only swiftly-obtained final decisions are of any value. In addition, dispute settlement clauses calling for final and binding decisions may have a dissuasive effect on the parties, as suggested, for example, by the dearth of disputes arbitrated under the ESA's arbitration provisions.[28]

Fourth, arbitral awards are internationally recognized and enforceable in all signatory states of the New York Convention, presently one hundred and forty-six.

Fifth, arbitral procedure is flexible and can be modified by agreement of the parties.[29] This may be of great use in the rapidly-evolving field of space activities.

Sixth, parties to arbitration choose their own decision-makers. Unlike in a court, where the only expertise parties can expect from the judge or judges assigned to their case is legal,

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[24] This proposal was made in the Final Draft of the Revised Convention on the Settlement of Disputes Related to Space Activities, *supra* note 6.

[25] *See* Outer Space Rules, *supra* note 1, at art. 34(2).

[26] *See* New York United Nations Convention on the Recognition and Enforcement of Foreign Arbitral Awards, art. V, June 7, 1959, 330 U.N.T.S. 38, 21 U.S.T. 2517, Article V [hereinafter New York Convention].

[27] GOH, *supra* note 9, at 116.

[28] Brisibe, Discussion Paper, *supra* note 5, at 6, citing André Farand, *The European Space Agency's Experience with Mechanisms for the Settlement of Disputes, in* INTERNATIONAL BUREAU OF THE PCA (ED.), ARBITRATION IN AIR, SPACE AND TELECOMMUNICATIONS LAW: ENFORCING REGULATORY MEASURES, THE PCA/PEACE PALACE PAPERS 145 (Kluwer Law International, 2002).

[29] *See* Outer Space Rules, *supra* note 1, at art. 1(1); WEBSTER, *supra* note 21, at 48-51.

parties in arbitration have the option of selecting arbitrators with specialized competences in the relevant fields. This is especially useful given the interdisciplinary nature of space activities, involving fields as diverse as economics, cutting-edge technology, and a tangle of scientific branches. On occasion, outer space technology such as remote sensing, may have legal or evidentiary limitations that are not immediately apparent to most adjudicators.[30]

Finally, arbitration can serve to preserve the confidentiality of sensitive information. Hearings need not be public and awards need not be published.[31] This is important for space-related disputes, as they may involve information concerning major state contracts and novel high technology, potentially treading the fine line between civil and military applications. The confidentiality of some of this information may be crucial to national security interests. For example, many states may insist on the sensitivity of remote sensing imagery.[32]

The Advisory Group, in drafting the Outer Space Rules, sought to further enhance many of these general features of international arbitration, as described below.

IV. DRAFTING THE RULES

A. *General Considerations*

The second part of the Advisory Group's mandate was the drafting of the Outer Space Rules. Here too, our methodology involved questionnaires and rounds of comments. This phase was marked by a higher degree of involvement by the PCA's International Bureau, and featured an in-person meeting of the Advisory Group in the Hague.

It is also worth mentioning that the work of the Advisory Group was punctuated by regular reports to the PCA Administrative Council, which keenly followed our progress. The initial draft of the Outer Space Rules was submitted for consideration

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[30] Williams, Discussion Paper, *supra* note 5, at 4-5.
[31] See Outer Space Rules, *supra* note 1, at arts. 28(3) & 34(5).
[32] Gabrynowicz, Discussion Paper, *supra* note 5, at 6-7.

by the PCA member states in May 2011. Throughout the summer of 2011, the member States sent us in depth commentary, which assisted us in ensuring that the Outer Space Rules would reflect, among others, the concerns of States and intergovernmental organizations. At the end of the process, the Administrative Council of the PCA adopted the Outer Space Rules by consensus.

Our strategy was to rely for the basis of our work on the 2010 UNCITRAL Rules, as well as on multiple sets of PCA procedural rules. The UNCITRAL Rules are the most widely used set of procedural rules in international commercial arbitration. They are an attractive model because their provisions have generated, since the adoption of their first version in 1976 by the UNCITRAL, an amount of case law and academic commentary much larger than that inspired by any other set of procedural rules for arbitration. By relying on the phrasing of the UNCITRAL Rules – whenever a departure from their provisions was not called for by some unique aspect of space-related disputes – we tapped into a wealth of precedent, thus enhancing the degree of predictability in the interpretation and application of the Outer Space Rules. Moreover, a new version of the UNCITRAL Rules was adopted in 2010, after protracted provision-by-provision revision discussions within the UNCITRAL Working Group II (Arbitration and Conciliation), taking into account many of the lessons learnt from thirty-four years of usage of the UNCITRAL Rules. We considered that the Outer Space Rules should benefit from these lessons. We did, however, find it equally beneficial to draw from other sources. Given the high proportion of participation by States, State agencies, and regional and international organizations in space activities, we found it most useful to seriously analyze the PCA rules of procedure tailored to use by such parties: the PCA *Optional Rules of Procedure for Arbitrating Disputes between Two States* (1992), the PCA *Optional Rules for Arbitrating Disputes between Two Parties of Which Only One is a State* (1993), the PCA *Optional Rules for Arbitration between International Organizations and States* (1996), and the PCA *Optional Rules for Arbitration of Disputes between International Organizations and Private Parties* (1996). Precious ideas and drafting were also borrowed from

the PCA's first set of sector specific rules, the PCA *Optional Rules for Arbitration of Disputes Relating to Natural Resources and/or the Environment*, aimed at the resolution of disputes concerning a subject matter – environment and natural resources – that shares momentous traits with outer space activities: among other, a high level of technical complexity and the sensitivity or confidentiality of information pertinent to the arbitral proceedings.

The result of the Advisory Group's efforts to adapt these models to the specificities of space-related disputes may best be illustrated by drawing attention to a few salient aspects of the Outer Space Rules.

B. *Specific Provisions*

In contrast to other dispute resolution instruments in international space law, the Outer Space Rules' scope of application is maximally broad. As is apparent from Article 1(1), the Outer Space Rules can be adopted by consent as the rules of procedure between any parties, whatever their nature. The final sentence of Article 1(1), providing that "the characterization of the dispute as relating to outer space is not necessary for jurisdiction," serves to manage the factual ambiguity that may arise over whether the dispute between the parties relates closely enough to "outer space."[33] While the Advisory Group considered drafting a test for determining whether or not a particular dispute was related to outer space, it was decided that where parties to a contract or other legal relationship agree to use the Outer Space Rules, the geographic, technological or other factual particularities of the dispute should not frustrate the parties' stated intent to proceed to arbitration.[34] The *ratione materiae* jurisdiction of the Outer Space Rules thus depends solely on the will of the parties and in no way on any conception of "outer space." Articles 3(3)(d) and 4(3)(d) further broaden the scope of the Outer Space Rules by enumerating among the documents

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[33] von der Dunk, Discussion Paper, *supra* note 5, at 2.

[34] Comments from Frans G. von der Dunk (Mar. 24, 2010) to the letter of Judge Pocar (Mar. 20,2010) (on file with PCA).

which must be identified in a Notice of Arbitration or a Response to the Notice of Arbitration the panoply of instruments to which space disputes may be related: "rule, decision, agreement, contract, convention, treaty, constituent instrument of an organization or agency." This language goes beyond that of the UNCITRAL Rules, in recognition of the variety of sources of law and the important role of States in space law.[35]

The involvement of States in space activities is further taken into account in Article 1(2) of the Outer Space Rules, which stipulates that agreement to arbitrate under the Outer Space Rules amounts to "a waiver of any right of immunity to jurisdiction." It is generally understood that consent to arbitration constitutes a waiver of immunity to jurisdiction.[36] This principle is made explicit in this article both as concerns the sovereign immunity of states and any immunity to jurisdiction that intergovernmental organizations may have.

The Outer Space Rules tackle the potential technical and scientific complexity of disputes relating to outer space activities from a variety of angles. As mentioned above, the possibility for parties to choose their decision-makers renders arbitration more attractive than adjudication for the settlement of space-related disputes. To enhance this advantage, Article 10(4) of the Outer Space Rules assists parties in their choice of arbitrators by mandating the Secretary-General of the PCA to compile a standing list of arbitrators with an expertise in space-related matters. The use of this list is optional. The qualifications of arbitrators are ensured because they are nominated by PCA member states and proposed by the Secretary-General, in consultation with the legal community in the relevant field.[37] Where its technical and scientific knowledge proves insufficient, the

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[35] *See e.g.* Frans G. von der Dunk, Response to First Questionnaire (Feb. 19, 2010); Ram Jakhu, Comments and Suggestions to First Questionnaire (Feb. 28, 2010); Joanne Gabrynowicz, Response to CP-OS 31655 and its Attached Annex (Feb. 28, 2010) (on file with the PCA).

[36] NIGEL BLACKABY, CONSTANTINE PARTASIDES ET AL., REDFERN AND HUNTER ON INTERNATIONAL ARBITRATION (Oxford, Oxford University Press, 5th ed., 2009), at para. 11.138.

[37] The list is currently being compiled and, once complete, will be made available on the PCA website at www.pca-cpa.org.

arbitral tribunal may appoint experts to assist it. Pursuant to Article 28(7) of the Outer Space Rules, the Secretary-General compiles a list of technical and scientific experts to facilitate the tribunal's choice of experts. As with the list of specially-qualified arbitrators, the use of this list is optional, leaving parties the option of selecting experts from such sources as the membership directory of the International Academy of Astronautics, which elects by secret ballot over a thousand leading experts in space and space-related science and technology.[38] Moreover, pursuant to Article 27(4) of the Outer Space Rules, the arbitral tribunal may enhance its understanding of technical issues by requesting the parties to provide a "non-technical document summarizing and explaining the background to any scientific, technical or other specialized information which the arbitral tribunal considers to be necessary to understand fully the matters in dispute." This type of document may assist the tribunal in assessing evidence and in determining whether experts need to be consulted. The provision for the possibility of a five-member tribunal, found at Article 9(1) of the Outer Space Rules, also anticipates the possibility of highly complex, high stake arbitrations, potentially involving crucial national State interests and large amounts in dispute.

Given the need for heightened confidentiality in space-related matters identified by the Advisory Group, the usual arbitral protections were expanded in the Outer Space Rules. Most significantly, Article 17(8) provides for the appointment of a "confidentiality adviser," whose role is to report to the tribunal on an issue on the basis of confidential information, without revealing the confidential content of the document to the tribunal or the other party. The technical justification for this mechanism is that the confidential information might be of such a technical nature that it would not mean much to the arbitrators, but would be meaningful to a confidentiality advisor who is also a technical expert.[39] The ethical justification for the confidenti-

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[38] *See* INTERNATIONAL ACADEMY OF ASTRONAUTICS, http://iaaweb.org (last visited June 15, 2012).

[39] Marc Blessing, *Arbitrability of Intellectual Property Disputes*, *in* 12:2 ARBITRATION INTERNATIONAL 191, 215 (Kluwer Law International, 1996).

ality advisor mechanism is that there may be times where the party would not wish the arbitrators to acquire knowledge of the confidential information, in particular "because of the fear that one party-nominated arbitrator might be an unscrupulous arbitrator who might be the source of a leak."[40]

The PCA has an active role under the Outer Space Rules. Article 1(3) provides for registry services and secretarial support by the PCA International Bureau. The PCA, because of its unique status as an intergovernmental organization with broad membership and its extensive experience managing arbitrations involving States or State entities,[41] is better positioned than private arbitral institutions to manage arbitrations involving the entire range of parties expected to be involved in outer space activities. Article 6(1) identifies the Secretary-General of the PCA as the default appointing authority under the Outer Space Rules. On this basis, the PCA Secretary-General has the responsibility, upon request by a party, to appoint, replace, and decide challenges against arbitrators.[42] While the option of naming the PCA Secretary-General as default appointing authority was extensively considered by the UNCITRAL Working Group charged with the drafting of the UNCITRAL Rules, it was not adopted, the PCA Secretary-General receiving instead the role of designating authority, charged with designating an appointing authority where none is agreed by the parties.[43] In contrast, by naming a specific appointing authority, the Outer Space Rules avoid potential delays in the constitution of the tribunal or during subsequent challenges to arbitrators for parties who have not previously agreed on the identity of an appointing authority.

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[40] *Id.*

[41] As of March 15, 2012, the PCA's docket includes 58 cases to which at least one party was a state or state entity.

[42] Outer Space Rules, *supra* note 1, at arts. 8(1), 9(2), 9(3), 10(3), 13(4), & 14(1).

[43] United Nations Commission on International Trade Law Arbitration Rules, at art. 6(1), Dec. 15, 1976, 15 I.L.M. 701; Report of the Working Group on Arbitration and Conciliation on the work of its forty-sixth Session, A/CN.9/619 at paras.71 & ff, (New York, Feb. 5-9, 2007).

V. CONCLUSION

In developing the Outer Space Rules, the Permanent Court of Arbitration and the Advisory Group sought to fill some of the fundamental *lacunae* in existing dispute resolution mechanisms of international space law. The rules provide a comprehensive voluntary dispute resolution procedure specifically tailored to the peculiarities of disputes relating to outer space activities. Their success depends entirely on how much confidence they can inspire in the international community. I am certain that the work and thought invested into the Outer Space Rules by the Advisory Group and the PCA International Bureau, described in this introduction, and the detail of the Rules' provisions, which I now invite you to explore, will inspire such confidence.

PERMANENT COURT OF ARBITRATION

OPTIONAL RULES
FOR ARBITRATION OF DISPUTES
RELATING TO OUTER SPACE ACTIVITIES

Effective December 6, 2011

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Introduction

These Rules are based on the 2010 UNCITRAL Arbitration Rules with changes in order to:

(i) reflect the particular characteristics of disputes having an outer space component involving the use of outer space by States, international organizations and private entities;

(ii) reflect the public international law element that pertains to disputes that may involve States and the use of outer space, and international practice appropriate to such disputes;

(iii) indicate the role of the Secretary-General and the International Bureau of the Permanent Court of Arbitration (PCA) at The Hague;

(iv) provide freedom for the parties to choose to have an arbitral tribunal of one, three or five persons;

(v) provide for establishment of a specialized list of arbitrators mentioned in article 10 and a list of scientific and technical experts mentioned in article 29 of these Rules; and

(vi) provide suggestions for establishing procedures aimed at ensuring confidentiality.

The Rules are optional and emphasize flexibility and party autonomy. For example:

(i) The Rules, and the services of the Secretary-General and the International Bureau of the PCA, are available to States, international organizations, and private parties; and

(ii) The Rules may be used, *inter alia*, in relation to disputes between two or more States parties to a multilateral agreement relating to the use of or access to outer space concerning the interpretation or application of that agreement.

Where arbitrations deal with technical questions, provision is made in article 27 for the submission to the arbitral tribunal of a document agreed to by the parties, summarizing and providing background to any scientific or technical issues that the parties may wish to raise in their memorials or at oral hearings.

A model clause that parties may consider inserting in treaties or other agreements to provide for arbitration of future disputes, and a model clause for arbitration of existing disputes are set forth in the annex to these Rules.

Section I. Introductory rules

Scope of application*

Article 1

1. Where parties have agreed that disputes between them in respect of a defined legal relationship, whether contractual or not, shall be referred to arbitration under the Permanent Court of Arbitration Optional Rules for Arbitration of Disputes Relating to Outer Space Activities, then such disputes shall be settled in accordance with these Rules subject to such modification as the parties may agree. The characterization of the dispute as relating to outer space is not necessary for jurisdiction where parties have agreed to settle a specific dispute under these Rules.

2. Agreement by a party to arbitration under these Rules constitutes a waiver of any right of immunity from jurisdiction, in respect of the dispute in question, to which such party might otherwise be entitled. A waiver of immunity relating to the execution of an arbitral award must be explicitly expressed.

3. The International Bureau of the Permanent Court of Arbitration (the 'International Bureau') shall serve as registry for the proceedings and provide secretariat services.

* A model arbitration clause for contracts can be found in the annex to the Rules.

Notice and calculation of periods of time

Article 2

1. A notice, including a notification, communication or proposal, may be transmitted by any means of communication that provides or allows for a record of its transmission.

2. If an address has been designated by a party specifically for this purpose or authorized by the arbitral tribunal any notice shall be delivered to that party at that address and if so delivered shall be deemed to have been received. Delivery by electronic means such as facsimile or email may only be made to an address so designated or authorized.

3. In the absence of such designation or authorization, a notice is:

   (a) received if it is physically delivered to the addressee; or

   (b) deemed to have been received if it is delivered at the place of business, habitual residence or mailing address of the addressee.

4. If, after reasonable efforts, delivery cannot be effected in accordance with paragraphs 2 or 3, a notice is deemed to have been received if it is sent to the addressee's last-known place of business, habitual residence or mailing address by registered letter or any other means that provides a record of delivery or of attempted delivery.

5. A notice shall be deemed to have been received on the day it is delivered in accordance with paragraphs 2, 3 or 4, or attempted to be delivered in accordance with paragraph 4. A notice transmitted by electronic means is deemed to have been received on the day it is sent, except that a

notice of arbitration so transmitted is only deemed to have been received on the day when it reaches the addressee's electronic address.

6. For the purpose of calculating a period of time under these Rules, such period shall begin to run on the day following the day when a notice is received. If the last day of such period is an official holiday or a non-business day at the residence or place of business of the addressee, the period is extended until the first business day which follows. Official holidays or non-business days occurring during the running of the period of time are included in calculating the period.

Notice of arbitration

Article 3

1. The party or parties initiating recourse to arbitration (hereinafter called the "claimant") shall communicate to the other party or parties (hereinafter called the "respondent") and the International Bureau a notice of arbitration.

2. Arbitral proceedings shall be deemed to commence on the date on which the notice of arbitration is received by the respondent.

3. The notice of arbitration shall include the following:

   (a) A demand that the dispute be referred to arbitration;

   (b) The names and contact details of the parties;

   (c) Identification of the arbitration agreement that is invoked;

   (d) Identification of any rule, decision, agreement, contract, convention, treaty, constituent instrument of an organization or agency, or relationship out of, or in relation to which, the dispute arises;

   (e) A brief description of the claim and an indication of the amount involved, if any;

   (f) The relief or remedy sought;

   (g) A proposal as to the number of arbitrators, language and place of arbitration, if the parties have not previously agreed thereon.

4. The notice of arbitration may also include:

   (a) A proposal for the appointment of a sole arbitrator referred to in article 8, paragraph 1;

   (b) Notification of the appointment of an arbitrator referred to in articles 9 or 10.

5. The constitution of the arbitral tribunal shall not be hindered by any controversy with respect to the sufficiency of the notice of arbitration, which shall be finally resolved by the arbitral tribunal.

Response to the notice of arbitration

Article 4

1. Within 30 days of the receipt of the notice of arbitration, the respondent shall communicate to the claimant and the International Bureau a response to the notice of arbitration, which shall include:

   (a) The name and contact details of each respondent;

   (b) A response to the information set forth in the notice of arbitration, pursuant to article 3, paragraphs 3 (c) to (g).

2. The response to the notice of arbitration may also include:

   (a) Any plea that an arbitral tribunal to be constituted under these Rules lacks jurisdiction;

   (b) A proposal for the appointment of a sole arbitrator referred to in article 8, paragraph 1;

   (c) Notification of the appointment of an arbitrator referred to in articles 9 or 10;

   (d) A brief description of counterclaims or claims for the purpose of a set-off, if any, including where relevant, an indication of the amounts involved, and the relief or remedy sought;

   (e) A notice of arbitration in accordance with article 3 in case the respondent formulates a claim against a party to the arbitration agreement other than the claimant.

3. The constitution of the arbitral tribunal shall not be hindered by any controversy with respect to the respondent's failure to communicate a response to the notice of arbitration, or an incomplete or late response to the notice of arbitration, which shall be finally resolved by the arbitral tribunal.

Representation and assistance

Article 5

Each party may be represented or assisted by persons chosen by it. The names and addresses of such persons must be communicated to all parties, to the International Bureau and to the arbitral tribunal. Such communication must specify whether the appointment is being made for purposes of representation or assistance. Where a person is to act as a representative of a party, the arbitral tribunal, on its own initiative or at the request of any party, may at any time require proof of authority granted to the representative in such a form as the arbitral tribunal may determine.

Appointing authority

Article 6

1. The Secretary-General of the PCA shall serve as appointing authority.

2. In exercising its functions under these Rules, the appointing authority may require from any party and the arbitrators the information it deems necessary and it shall give the parties and, where appropriate, the arbitrators, an opportunity to present their views in any manner it considers

appropriate. All such communications to and from the appointing authority shall also be provided by the sender to all other parties.

3. The appointing authority shall have regard to such considerations as are likely to secure the appointment of an independent and impartial arbitrator and shall take into account the advisability of appointing an arbitrator of a nationality other than the nationalities of the parties.

Section II. Composition of the arbitral tribunal

Number of arbitrators

Article 7

1. If the parties have not previously agreed on the number of arbitrators, and if within 30 days after the receipt by the respondent of the notice of arbitration the parties have not agreed that there shall be only one arbitrator, three arbitrators shall be appointed.

2. Notwithstanding paragraph 1, if no other parties have responded to a party's proposal to appoint a sole arbitrator within the time limit provided for in paragraph 1 and the party or parties concerned have failed to appoint a second arbitrator in accordance with articles 9 or 10, the appointing authority may, at the request of a party, appoint a sole arbitrator pursuant to the procedure provided for in article 8, paragraph 2 if it determines that, in view of the circumstances of the case, this is more appropriate.

Appointment of arbitrators (articles 8 to 10)

Article 8

1. If the parties have agreed that a sole arbitrator is to be appointed and if within 30 days after receipt by all other parties of a proposal for the appointment of a sole arbitrator the parties have not reached agreement thereon, a sole arbitrator shall, at the request of a party, be appointed by the appointing authority.

2. The appointing authority shall appoint the sole arbitrator as promptly as possible. In making the appointment, the appointing authority shall use the following list-procedure, unless the parties agree that the list-procedure should not be used or unless the appointing authority determines in its discretion that the use of the list-procedure is not appropriate for the case:

   (a) The appointing authority shall communicate to each of the parties an identical list containing at least three names;

   (b) Within 15 days after the receipt of this list, each party may return the list to the appointing authority after having deleted the name or names to which it objects and numbered the remaining names on the list in the order of its preference;

   (c) After the expiration of the above period of time the appointing authority shall appoint the sole arbitrator from among the names approved on the lists returned to it and in accordance with the order of preference indicated by the parties;

   (d) If for any reason the appointment cannot be made according to this procedure, the appointing authority may exercise its discretion in appointing the sole arbitrator.

Article 9

1. If three arbitrators are to be appointed, each party shall appoint one arbitrator. The two arbitrators thus appointed shall choose the third arbitrator who will act as the presiding arbitrator of the arbitral tribunal. If five arbitrators are to be appointed, the two party-appointed arbitrators shall choose the remaining three arbitrators and designate one of those three as the presiding arbitrator of the tribunal.

2. If within 30 days after the receipt of a party's notification of the appointment of an arbitrator the other party has not notified the first party of the arbitrator it has appointed, the first party may request the appointing authority to appoint the second arbitrator.

3. If within 30 days after the appointment of the second arbitrator the two arbitrators have not agreed on the choice of the remaining arbitrators and/or the presiding arbitrator, the remaining arbitrators and/or the presiding arbitrator shall be appointed by the appointing authority in the same way as a sole arbitrator would be appointed under article 8.

Article 10

1. For the purposes of article 9, paragraph 1, where three or five arbitrators are to be appointed and there are multiple parties as claimant or as respondent, unless the parties have agreed to another method of appointment of arbitrators, the multiple parties jointly, whether as claimant or as respondent, shall appoint an arbitrator.

2. If the parties have agreed that the arbitral tribunal is to be composed of a number of arbitrators other than one, three, or five, the arbitrators shall be appointed according to the method agreed upon by the parties.

3. In the event of any failure to constitute the arbitral tribunal under these Rules, the appointing authority shall, at the request of any party, constitute the arbitral tribunal and, in doing so, may revoke any appointment already made and appoint or reappoint each of the arbitrators and designate one of them as the presiding arbitrator.

4. In appointing arbitrators pursuant to these Rules, the parties and the appointing authority are free to designate persons who are not Members of the Permanent Court of Arbitration at The Hague. For the purpose of assisting the parties the Secretary-General will make available a list of persons considered to have expertise in the subject matters of the dispute at hand for which these Rules have been designed.

Disclosures by and challenge of arbitrators (articles 11 to 13)**

Article 11

When a person is approached in connection with his or her possible appointment as an arbitrator, he or she shall disclose any circumstances likely to give rise to justifiable doubts as to his or her impartiality or independence. An arbitrator, from the time of his or her appointment and throughout the arbitral proceedings, shall without delay disclose any such circumstances to the parties and the other arbitrators unless they have already been informed by him or her of these circumstances.

** Model statements of independence pursuant to article 11 can be found in the annex to the Rules.

Article 12

1. Any arbitrator may be challenged if circumstances exist that give rise to justifiable doubts as to the arbitrator's impartiality or independence or if he or she does not have the qualifications agreed by the parties in their arbitration agreement.

2. A party may challenge the arbitrator appointed by it only for reasons of which it becomes aware after the appointment has been made.

3. In the event that an arbitrator fails to act or in the event of the *de jure* or *de facto* impossibility of his or her performing his or her functions, the procedure in respect of the challenge of an arbitrator as provided in article 13 shall apply.

4. If an arbitrator on a three- or five-person tribunal fails to participate in the arbitration, the other arbitrators shall, unless the parties agree otherwise, have the power in their sole discretion to continue the arbitration and to make any decision, ruling or award, notwithstanding the failure of one arbitrator to participate. In determining whether to continue the arbitration or to render any decision, ruling or award without the participation of an arbitrator, the other arbitrators shall take into account the stage of the arbitration, the reason, if any, expressed by the arbitrator for such non-participation, and such other matters as they consider appropriate in the circumstances of the case. In the event that the other arbitrators determine not to continue the arbitration without the non-participating arbitrator, the arbitral tribunal shall declare the office vacant, and, subject to article 14(2), a substitute arbitrator shall be appointed pursuant to the provisions of articles 8 to 11.

Article 13

1. A party that intends to challenge an arbitrator shall send notice of its challenge within 30 days after it has been notified of the appointment of the challenged arbitrator, or within 30 days after the circumstances mentioned in articles 11 and 12 became known to that party.

2. The notice of challenge shall be communicated to all other parties, to the arbitrator who is challenged and to the other arbitrators. The notice of challenge shall state the reasons for the challenge.

3. When an arbitrator has been challenged by a party, all parties may agree to the challenge. The arbitrator may also, after the challenge, withdraw from his or her office. In neither case does this imply acceptance of the validity of the grounds for the challenge.

4. If, within 15 days from the date of the notice of challenge, all parties do not agree to the challenge or the challenged arbitrator does not withdraw, the party making the challenge may elect to pursue it. In that case, within 30 days from the date of the notice of challenge, it shall seek a decision on the challenge by the appointing authority.

Replacement of an arbitrator

Article 14

1. Subject to paragraph 2, in any event where an arbitrator has to be replaced during the course of the arbitral proceedings, a substitute arbitrator shall be appointed or chosen pursuant to the procedure provided for in articles 8 to 11 that was applicable to the appointment or choice of the arbitrator being replaced. This procedure shall apply even if during the process of appointing the arbitrator to be replaced, a party had failed to exercise its right to appoint or to participate in the appointment.

2. If, at the request of a party, the appointing authority determines that, in view of the exceptional circumstances of the case, it would be justified for a party to be deprived of its right to appoint a substitute arbitrator, the appointing authority may, after giving an opportunity to the parties and the remaining arbitrators to express their views, appoint the substitute arbitrator.

Repetition of hearings in the event of the replacement of an arbitrator

Article 15

If an arbitrator is replaced, the proceedings shall resume at the stage where the arbitrator who was replaced ceased to perform his or her functions, unless the arbitral tribunal decides otherwise.

Exclusion of liability

Article 16

The parties waive, to the fullest extent permitted under the applicable law, any claim against the arbitrators and any person appointed by the arbitral tribunal based on any act or omission in connection with the arbitration.

Section III. Arbitral proceedings

General provisions

Article 17

1. Subject to these Rules, the arbitral tribunal may conduct the arbitration in such manner as it considers appropriate, provided that the parties are treated with equality and that at an appropriate stage of the proceedings each party is given a reasonable opportunity of presenting its case. The arbitral tribunal, in exercising its discretion, shall conduct the proceedings so as to avoid unnecessary delay and expense and to provide a fair and efficient process for resolving the parties' dispute.

2. As soon as practicable after its constitution and after inviting the parties to express their views, the arbitral tribunal shall establish the provisional timetable of the arbitration. The arbitral tribunal may, at any time, after inviting the parties to express their views, extend or abridge any period of time prescribed under these Rules or agreed by the parties.

3. If at an appropriate stage of the proceedings any party so requests, the arbitral tribunal shall hold hearings for the presentation of evidence by witnesses, including expert witnesses, or for oral argument. In the absence of such a request, the arbitral tribunal shall decide whether to hold such

hearings or whether the proceedings shall be conducted on the basis of documents and other materials.

4. All communications to the arbitral tribunal by one party shall be communicated by that party to all other parties and the International Bureau. Such communications shall be made at the same time, except as otherwise permitted by the arbitral tribunal if it may do so under applicable law.

5. The arbitral tribunal may, at the request of any party, allow one or more third persons to be joined in the arbitration as a party provided such person is a party to the arbitration agreement, unless the arbitral tribunal finds, after giving all parties, including the person or persons to be joined, the opportunity to be heard, that joinder should not be permitted because of prejudice to any of those parties. The arbitral tribunal may make a single award or several awards in respect of all parties so involved in the arbitration.

6. A party invoking the confidentiality of any information it wishes or is required to submit in the arbitration, including to an expert appointed by the arbitral tribunal, shall make an application to have the information classified as confidential by notice containing the reasons for which it considers the information confidential to the arbitral tribunal, with a copy to the other party and the International Bureau.

7. The arbitral tribunal shall determine whether the information is to be classified as confidential and of such a nature that the absence of special measures of protection in the proceedings would be likely to cause serious harm to the party or parties invoking its confidentiality. If the arbitral tribunal so determines, it shall decide and communicate in writing to the parties and the International Bureau under what conditions and to whom the confidential information may in part or in whole be disclosed and shall require any person to whom the confidential information is to be disclosed to sign an appropriate confidentiality undertaking.

8. The arbitral tribunal may also, at the request of a party or on its own motion, appoint a confidentiality adviser as an expert in accordance with article 29 in order to report to it on the basis of the confidential information on specific issues designated by the arbitral tribunal without disclosing the confidential information either to the party from whom the confidential information does not originate or to the arbitral tribunal.

Place of arbitration

Article 18

1. If the parties have not previously agreed on the place of arbitration, the place of arbitration shall be determined by the arbitral tribunal having regard to the circumstances of the case. The award shall be deemed to have been made at the place of arbitration.

2. The arbitral tribunal may meet at any location it considers appropriate for deliberations. Unless otherwise agreed by the parties, the arbitral tribunal may also meet at any location it considers appropriate for any other purpose, including hearings.

Language

Article 19

1. Subject to an agreement by the parties, the arbitral tribunal shall, promptly after its appointment, determine the language or languages to be used in the proceedings. This determination shall apply to the statement of claim, the statement of defence, and any further written statements and, if oral hearings take place, to the language or languages to be used in such hearings.

2. The arbitral tribunal may order that any documents annexed to the statement of claim or statement of defence, and any supplementary documents or exhibits submitted in the course of the proceedings, delivered in their original language, shall be accompanied by a translation into the language or languages agreed upon by the parties or determined by the arbitral tribunal.

Statement of claim

Article 20

1. The claimant shall communicate its statement of claim in writing to the respondent, to the International Bureau, and to each of the arbitrators within a period of time to be determined by the arbitral tribunal. The claimant may elect to treat its notice of arbitration referred to in article 3 as a statement of claim, provided that the notice of arbitration also complies with the requirements of paragraphs 2 to 4 of this article.

2. The statement of claim shall include the following particulars:

   (a) The names and contact details of the parties;

   (b) A statement of the facts supporting the claim;

   (c) The points at issue;

   (d) The relief or remedy sought;

   (e) The legal grounds or arguments supporting the claim.

3. A copy of any rule, decision, agreement, contract, convention, treaty, constituent instrument of an organization or agency, or relationship out of, or in relation to which, the dispute arises and of the arbitration agreement shall be annexed to the statement of claim.

4. The statement of claim should, as far as possible, be accompanied by all documents and other evidence relied upon by the claimant, or contain references to them.

Statement of defence

Article 21

1. The respondent shall communicate its statement of defence in writing to the claimant, to the International Bureau, and to each of the arbitrators within a period of time to be determined by the arbitral tribunal. The respondent may elect to treat its response to the notice of arbitration referred to

in article 4 as a statement of defence, provided that the response to the notice of arbitration also complies with the requirements of paragraph 2 of this article.

2. The statement of defence shall reply to the particulars (b) to (e) of the statement of claim (article 20, paragraph 2). The statement of defence should, as far as possible, be accompanied by all documents and other evidence relied upon by the respondent, or contain references to them.

3. In its statement of defence, or at a later stage in the arbitral proceedings if the arbitral tribunal decides that the delay was justified under the circumstances, the respondent may make a counterclaim or rely on a claim for the purpose of a set-off provided that the arbitral tribunal has jurisdiction over it.

4. The provisions of article 20, paragraphs 2 to 4 shall apply to a counterclaim, a claim under article 4, paragraph (2)(e) and a claim relied on for the purpose of a set-off.

Amendments to the claim or defence

Article 22

During the course of the arbitral proceedings, a party may amend or supplement its claim or defence, including a counterclaim or a claim for the purpose of a set-off, unless the arbitral tribunal considers it inappropriate to allow such amendment or supplement having regard to the delay in making it or prejudice to other parties or any other circumstances. However, a claim or defence, including a counterclaim or a claim for the purpose of a set-off, may not be amended or supplemented in such a manner that the amended or supplemented claim or defence falls outside the jurisdiction of the arbitral tribunal.

Pleas as to the jurisdiction of the arbitral tribunal

Article 23

1. The arbitral tribunal shall have the power to rule on its own jurisdiction, including any objections with respect to the existence or validity of the arbitration agreement. For that purpose, an arbitration clause that forms part of a contract shall be treated as an agreement independent of the other terms of the contract. A decision by the arbitral tribunal that the contract is null shall not entail automatically the invalidity of the arbitration clause.

2. A plea that the arbitral tribunal does not have jurisdiction shall be raised no later than in the statement of defence or, with respect to a counterclaim or a claim for the purpose of a set-off, in the reply to the counterclaim or to the claim for the purpose of a set-off. A party is not precluded from raising such a plea by the fact that it has appointed, or participated in the appointment of, an arbitrator. A plea that the arbitral tribunal is exceeding the scope of its authority shall be raised as soon as the matter alleged to be beyond the scope of its authority is raised during the arbitral proceedings. The arbitral tribunal may, in either case, admit a later plea if it considers the delay justified.

3. The arbitral tribunal may rule on a plea referred to in paragraph 2 either as a preliminary question or in an award on the merits. The arbitral tribunal may continue the arbitral proceedings and make an award, notwithstanding any pending challenge to its jurisdiction before a court.

Further written statements

Article 24

The arbitral tribunal shall decide which further written statements, in addition to the statement of claim and the statement of defence, shall be required from the parties or may be presented by them and shall fix the periods of time for communicating such statements.

Periods of time

Article 25

The periods of time fixed by the arbitral tribunal for the communication of written statements (including the statement of claim and statement of defence) should not exceed 45 days. However, the arbitral tribunal may extend the time limits if it concludes that an extension is justified.

Interim measures

Article 26

1. The arbitral tribunal may, at the request of a party, grant interim measures.

2. An interim measure is any temporary measure by which, at any time prior to the issuance of the award by which the dispute is finally decided, the arbitral tribunal orders a party, for example and without limitation, to:

   (a) Maintain or restore the status quo pending determination of the dispute;

   (b) Take action that would prevent, or refrain from taking action that is likely to cause, (i) current or imminent harm or (ii) prejudice to the arbitral process itself;

   (c) Provide a means of preserving assets out of which a subsequent award may be satisfied; or

   (d) Preserve evidence that may be relevant and material to the resolution of the dispute.

3. The party requesting an interim measure under paragraphs 2 (a) to (c) shall satisfy the arbitral tribunal that:

   (a) Harm not adequately reparable by an award of damages is likely to result if the measure is not ordered, and such harm substantially outweighs the harm that is likely to result to the party against whom the measure is directed if the measure is granted; and

   (b) There is a reasonable possibility that the requesting party will succeed on the merits of the claim. The determination on this possibility shall not affect the discretion of the arbitral tribunal in making any subsequent determination.

4. With regard to a request for an interim measure under paragraph 2 (d), the requirements in paragraphs 3 (a) and (b) shall apply only to the extent the arbitral tribunal considers appropriate.

5. The arbitral tribunal may modify, suspend or terminate an interim measure it has granted, upon application of any party or, in exceptional circumstances and upon prior notice to the parties, on the arbitral tribunal's own initiative.

6. The arbitral tribunal may require the party requesting an interim measure to provide appropriate security in connection with the measure.

7. The arbitral tribunal may require any party promptly to disclose any material change in the circumstances on the basis of which the interim measure was requested or granted.

8. The party requesting an interim measure may be liable for any costs and damages caused by the measure to any party if the arbitral tribunal later determines that, in the circumstances then prevailing, the measure should not have been granted. The arbitral tribunal may award such costs and damages at any point during the proceedings.

9. A request for interim measures addressed by any party to a judicial authority shall not be deemed incompatible with the agreement to arbitrate, or as a waiver of that agreement.

Evidence

Article 27

1. Each party shall have the burden of proving the facts relied on to support its claim or defence.

2. Witnesses, including expert witnesses, who are presented by the parties to testify to the arbitral tribunal on any issue of fact or expertise may be any individual, notwithstanding that the individual is a party to the arbitration or in any way related to a party. Unless otherwise directed by the arbitral tribunal, statements by witnesses, including expert witnesses, may be presented in writing and signed by them.

3. At any time during the arbitral proceedings the arbitral tribunal may require the parties to produce documents, exhibits or other evidence within such a period of time as the arbitral tribunal shall determine.

4. The arbitral tribunal may request the parties jointly or separately to provide a non-technical document summarizing and explaining the background to any scientific, technical or other specialized information which the arbitral tribunal considers to be necessary to understand fully the matters in dispute.

5. The arbitral tribunal shall determine the admissibility, relevance, materiality and weight of the evidence offered.

Hearings

Article 28

1. In the event of an oral hearing, the arbitral tribunal shall give the parties adequate advance notice of the date, time and place thereof.

2. Witnesses, including expert witnesses, may be heard under the conditions and examined in the manner set by the arbitral tribunal.

3. Hearings shall be held *in camera* unless the parties agree otherwise. The arbitral tribunal may require the retirement of any witness or witnesses, including expert witnesses, during the testimony of such other witnesses, except that a witness, including an expert witness, who is a party to the arbitration shall not, in principle, be asked to retire.

4. The arbitral tribunal may direct that witnesses, including expert witnesses, be examined through means of telecommunication that do not require their physical presence at the hearing (such as videoconference).

Experts appointed by the arbitral tribunal

Article 29

1. After consultation with the parties, the arbitral tribunal may appoint one or more independent experts to report to it, in writing, on specific issues to be determined by the arbitral tribunal. A copy of the expert's terms of reference, established by the arbitral tribunal, shall be communicated to the parties.

2. The expert shall, in principle before accepting appointment, submit to the arbitral tribunal and to the parties a description of his or her qualifications and a statement of his or her impartiality and independence. Within the time ordered by the arbitral tribunal, the parties shall inform the arbitral tribunal whether they have any objections as to the expert's qualifications, impartiality or independence. The arbitral tribunal shall decide promptly whether to accept any such objections.

3. After an expert's appointment, a party may object to the expert's qualifications, impartiality or independence only if the objection is for reasons of which the party becomes aware after the appointment has been made. The arbitral tribunal shall decide promptly what, if any, action to take.

4. The parties shall give the expert any relevant information or produce for his or her inspection any relevant documents or goods that he or she may require of them, subject to the provisions for confidentiality in article 17, paragraphs 6 to 8. Any dispute between a party and such expert as to the relevance of the required information or production shall be referred to the arbitral tribunal for decision.

5. Upon receipt of the expert's report, the arbitral tribunal shall communicate a copy of the report to the parties, which shall be given the opportunity to express, in writing, their opinion on the report. A party shall be entitled to examine any document on which the expert has relied in his or her report, subject to the provisions for confidentiality in article 17, paragraphs 6 and 7.

6. At the request of any party, the expert, after delivery of the report, may be heard at a hearing where the parties shall have the opportunity to be present and to interrogate the expert. At this hearing, any party may present expert witnesses in order to testify on the points at issue. The provisions of article 28 shall be applicable to such proceedings.

7. The Secretary-General will provide an indicative list of persons considered to have expertise in the scientific or technical matters in respect of which these Rules might be relied upon. In appointing one or more experts pursuant to paragraph 1 above, the arbitral tribunal shall not be limited in its choice to any person or persons appearing on the indicative list of experts.

Default

Article 30

1. If, within the period of time fixed by these Rules or the arbitral tribunal, without showing sufficient cause:

(a) The claimant has failed to communicate its statement of claim, the arbitral tribunal shall issue an order for the termination of the arbitral proceedings, unless there are remaining matters that may need to be decided and the arbitral tribunal considers it appropriate to do so;

(b) The respondent has failed to communicate its response to the notice of arbitration or its statement of defence, the arbitral tribunal shall order that the proceedings continue, without treating such failure in itself as an admission of the claimant's allegations; the provisions of this subparagraph also apply to a claimant's failure to submit a defence to a counterclaim or to a claim for the purpose of a set-off.

2. If a party, duly notified under these Rules, fails to appear at a hearing, without showing sufficient cause for such failure, the arbitral tribunal may proceed with the arbitration.

3. If a party, duly invited by the arbitral tribunal to produce documents, exhibits or other evidence, fails to do so within the established period of time, without showing sufficient cause for such failure, the arbitral tribunal may make the award on the evidence before it.

Closure of hearings

Article 31

1. The arbitral tribunal may inquire of the parties if they have any further proof to offer or witnesses to be heard or submissions to make and, if there are none, it may declare the hearings closed.

2. The arbitral tribunal may, if it considers it necessary owing to exceptional circumstances, decide, on its own initiative or upon application of a party, to reopen the hearings at any time before the award is made.

Waiver of right to object

Article 32

A failure by any party to object promptly to any noncompliance with these Rules or with any requirement of the arbitration agreement shall be deemed to be a waiver of the right of such party to make such an objection, unless such party can show that, under the circumstances, its failure to object was justified.

Section IV. The award

Decisions

Article 33

1. When there is more than one arbitrator, any award or other decision of the arbitral tribunal shall be made by a majority of the arbitrators.

2. In the case of questions of procedure, when there is no majority or when the arbitral tribunal so authorizes, the presiding arbitrator may decide alone, subject to revision, if any, by the arbitral tribunal.

Form and effect of the award

Article 34

1. The arbitral tribunal may make separate awards on different issues at different times.

2. All awards shall be made in writing and shall be final and binding on the parties. The parties shall carry out all awards without delay.

3. The arbitral tribunal shall state the reasons upon which the award is based, unless the parties have agreed that no reasons are to be given.

4. An award shall be signed by the arbitrators and it shall contain the date on which the award was made and indicate the place of arbitration. Where there is more than one arbitrator and any of them fails to sign, the award shall state the reason for the absence of the signature.

5. An award may be made public with the consent of all parties or where and to the extent disclosure is required of a party by legal duty, to protect or pursue a legal right or in relation to legal proceedings before a court or other competent authority.

6. Copies of the award signed by the arbitrators shall be communicated to the parties by the International Bureau.

7. Separate or dissenting opinions (if any) shall be in writing and signed by the dissenting arbitrator or arbitrators.

Applicable law, *amiable compositeur*

Article 35

1. In resolving the dispute, the arbitral tribunal shall apply the law or rules of law designated by the parties as applicable to the substance of the dispute. Failing such designation by the parties, the arbitral tribunal shall apply the national and/or international law and rules of law it determines to be appropriate.

2. The arbitral tribunal shall decide as *amiable compositeur* or *ex aequo et bono* only if the parties have expressly authorized the arbitral tribunal to do so.

3. In all cases, the arbitral tribunal shall decide in accordance with the terms of the contract, if any, and shall take into account any usage of trade applicable to the transaction.

Settlement or other grounds for termination

Article 36

1. If, before the award is made, the parties agree on a settlement of the dispute, the arbitral tribunal shall either issue an order for the termination of the arbitral proceedings or, if requested by the parties and accepted by the arbitral tribunal, record the settlement in the form of an arbitral award on agreed terms. The arbitral tribunal is not obliged to give reasons for such an award.

2. If, before the award is made, the continuation of the arbitral proceedings becomes unnecessary or impossible for any reason not mentioned in paragraph 1, the arbitral tribunal shall inform the parties of its intention to issue an order for the termination of the proceedings. The arbitral tribunal shall have the power to issue such an order unless there are remaining matters that may need to be decided and the arbitral tribunal considers it appropriate to do so.

3. Copies of the order for termination of the arbitral proceedings or of the arbitral award on agreed terms, signed by the arbitrators, shall be communicated by the arbitral tribunal to the parties. Where an arbitral award on agreed terms is made, the provisions of article 34, paragraphs 2, 4 and 5 shall apply.

Interpretation of the award

Article 37

1. Within 30 days after the receipt of the award, a party, with notice to the other parties and the International Bureau, may request that the arbitral tribunal give an interpretation of the award.

2. The interpretation shall be given in writing within 45 days after the receipt of the request. The interpretation shall form part of the award and the provisions of article 34, paragraphs 2 to 6, shall apply.

Correction of the award

Article 38

1. Within 30 days after the receipt of the award, a party, with notice to the other parties and the International Bureau, may request the arbitral tribunal to correct in the award any error in computation, any clerical or typographical error, or any error or omission of a similar nature. If the arbitral tribunal considers that the request is justified, it shall make the correction within 45 days of receipt of the request.

2. The arbitral tribunal may within 30 days after the communication of the award make such corrections on its own initiative.

3. Such corrections shall be in writing and shall form part of the award. The provisions of article 34, paragraphs 2 to 6, shall apply.

Additional award

Article 39

1. Within 30 days after the receipt of the termination order or the award, a party, with notice to the other parties and the International Bureau, may request the arbitral tribunal to make an award or an additional award as to claims presented in the arbitral proceedings but not decided by the arbitral tribunal.

2. If the arbitral tribunal considers the request for an award or additional award to be justified, it shall render or complete its award within 60 days after the receipt of the request. The arbitral tribunal may extend, if necessary, the period of time within which it shall make the award.

3. When such an award or additional award is made, the provisions of article 34, paragraphs 2 to 6, shall apply.

Definition of costs

Article 40

1. The arbitral tribunal shall fix the costs of arbitration in the final award and, if it deems appropriate, in another decision.

2. The term "costs" includes only:

   (a) The fees of the arbitral tribunal to be stated separately as to each arbitrator and to be fixed by the tribunal itself in accordance with article 41;

   (b) The reasonable travel and other expenses incurred by the arbitrators;

   (c) The reasonable costs of expert advice and of other assistance required by the arbitral tribunal;

   (d) The reasonable travel and other expenses of witnesses to the extent such expenses are approved by the arbitral tribunal;

   (e) The legal and other costs incurred by the parties in relation to the arbitration to the extent that the arbitral tribunal determines that the amount of such costs is reasonable;

   (f) The fees and expenses of the International Bureau, including the fees and expenses of the appointing authority.

3. In relation to interpretation, correction or completion of any award under articles 37 to 39, the arbitral tribunal may charge the costs referred to in paragraphs 2 (b) to (f), but no additional fees.

Fees and expenses of arbitrators

Article 41

1. The fees and expenses of the arbitrators shall be reasonable in amount, taking into account the amount in dispute, the complexity of the subject matter, the time spent by the arbitrators and any other relevant circumstances of the case.

2. Promptly after its constitution, the arbitral tribunal shall inform the parties as to how it proposes to determine its fees and expenses, including any rates it intends to apply. Within 15 days of receiving that proposal, any party may refer the proposal to the appointing authority for review. If, within 45 days of receipt of such a referral, the appointing authority finds that the proposal of the arbitral tribunal is inconsistent with paragraph 1, it shall make any necessary adjustments thereto, which shall be binding upon the arbitral tribunal.

3.
   (a) When informing the parties of the arbitrators' fees and expenses that have been fixed pursuant to article 40, paragraphs 2 (a) and (b), the arbitral tribunal shall also explain the manner in which the corresponding amounts have been calculated;

   (b) Within 15 days of receiving the arbitral tribunal's determination of fees and expenses, any party may refer for review such determination to the appointing authority;

   (c) If the appointing authority finds that the arbitral tribunal's determination is inconsistent with the arbitral tribunal's proposal (and any adjustment thereto) under paragraph 2 or is otherwise manifestly excessive, it shall, within 45 days of receiving such a referral, make any adjustments to the arbitral tribunal's determination that are necessary to satisfy the criteria in paragraph 1. Any such adjustments shall be binding upon the arbitral tribunal;

   (d) Any such adjustments shall either be included by the arbitral tribunal in its award or, if the award has already been issued, be implemented in a correction to the award, to which the procedure of article 38, paragraph 3 shall apply.

4. Throughout the procedure under paragraphs 2 and 3, the arbitral tribunal shall proceed with the arbitration, in accordance with article 17, paragraph 1.

5. A referral under paragraph 3 shall not affect any determination in the award other than the arbitral tribunal's fees and expenses; nor shall it delay the recognition and enforcement of all parts of the award other than those relating to the determination of the arbitral tribunal's fees and expenses.

Allocation of costs

Article 42

1. The costs of the arbitration shall in principle be borne by the unsuccessful party or parties. However, the arbitral tribunal may apportion each of such costs between the parties if it determines that apportionment is reasonable, taking into account the circumstances of the case.

2. The arbitral tribunal shall in the final award or, if it deems appropriate, in any other award, determine any amount that a party may have to pay to another party as a result of the decision on allocation of costs.

Deposit of costs

Article 43

1. The International Bureau, following the commencement of the arbitration, may request the parties to deposit an equal amount as an advance for the costs referred to in article 40, paragraphs 2(a), (b), (c), and (f). All amounts deposited by the parties pursuant to this paragraph 1 and paragraph 2 of this article shall be directed to the International Bureau, and disbursed by it for such costs, including, *inter alia*, fees to the arbitrators, the appointing authority, and the International Bureau.

2. Security for the costs of interim measures shall be directed to the International Bureau and disbursed by it upon order from the arbitral tribunal.

3. During the course of the arbitral proceedings the International Bureau may request supplementary deposits from the parties.

4. If the requested deposits are not paid in full within 60 days after the receipt of the request, the International Bureau shall so inform the parties in order that one or more of them may make the required payment. If such payment is not made, the arbitral tribunal may order the suspension or termination of the arbitral proceedings.

5. After a termination order or final award has been made, the International Bureau shall render an accounting to the parties of the deposits received and return any unexpended balance to the parties.

Annex

Model arbitration clause for contracts

Any dispute, controversy or claim arising out of or relating to this contract, or the breach, termination or invalidity thereof, shall be settled by arbitration in accordance with the PCA Optional Rules for Arbitration of Disputes Relating to Outer Space Activities.

Note — Parties should consider adding:

(a) The number of arbitrators shall be ... (one, three or five);
(b) The place of arbitration shall be ... (town and country);
(c) The language to be used in the arbitral proceedings shall be ... .

Possible waiver statement

Note — If the parties wish to exclude recourse against the arbitral award that may be available under the applicable law, they may consider adding a provision to that effect as suggested below, considering, however, that the effectiveness and conditions of such an exclusion depend on the applicable law.

Waiver: The parties hereby waive their right to any form of recourse against an award to any court or other competent authority, insofar as such waiver can validly be made under the applicable law.

Model statements of independence pursuant to article 11 of the Rules

No circumstances to disclose: I am impartial and independent of each of the parties and intend to remain so. To the best of my knowledge, there are no circumstances, past or present, likely to give rise to justifiable doubts as to my impartiality or independence. I shall promptly notify the parties and the other arbitrators of any such circumstances that may subsequently come to my attention during this arbitration.

Circumstances to disclose: I am impartial and independent of each of the parties and intend to remain so. Attached is a statement made pursuant to article 11 of the PCA Optional Rules for Arbitration of Disputes Relating to Outer Space Activities of (a) my past and present professional, business and other relationships with the parties and (b) any other relevant circumstances. [Include statement] I confirm that those circumstances do not affect my independence and impartiality. I shall promptly notify the parties and the other arbitrators of any such further relationships or circumstances that may subsequently come to my attention during this arbitration.

Note — Any party may consider requesting from the arbitrator the following addition to the statement of independence:

I confirm, on the basis of the information presently available to me, that I can devote the time necessary to conduct this arbitration diligently, efficiently and in accordance with the time limits in the Rules.